8-Puzzle

Problem model, counting conventions and report scope

STATE

Nine tile values in row order. 0 denotes the blank; every numbered tile 1-8 occurs once.

INITIAL STATE

1 2 3 / 0 4 6 / 7 5 8. This is the start board supplied by the user.

ACTIONS

Move the blank Up, Down, Left or Right, in that order. Only on-board moves are generated.

SEARCH RESULT

Visited / dequeued states: 17
Unique discovered, including the initial state: 30
Attempted transitions: 44
Rejected invalid / repeated: 0 / 15
Shortest solution depth: 3

CONSTRAINTS

Swap the blank with one orthogonally adjacent tile. Preserve all nine tile values.

GOAL

1 2 3 / 4 5 6 / 7 8 0.

STRATEGY

Unit-cost BFS with a global discovered-state set. Test the goal when removing a state from the FIFO queue.

PAGES THAT FOLLOW

The next 17 pages show every BFS expansion or goal check up to the first dequeued goal, followed by the final route. Every page retains the tree generated so far, starting at the root, and enlarges the current expansion below. Repeated states are yellow attempts and never enter the queue twice. The overview uses fixed positions; zoom the vector PDF to inspect small nodes.

COUNTING NOTE

Statistics apply only to this instance, action order and goal-test timing. Attempted-transition nodes are not the same as unique discovered states. Green nodes identify the returned route retrospectively.

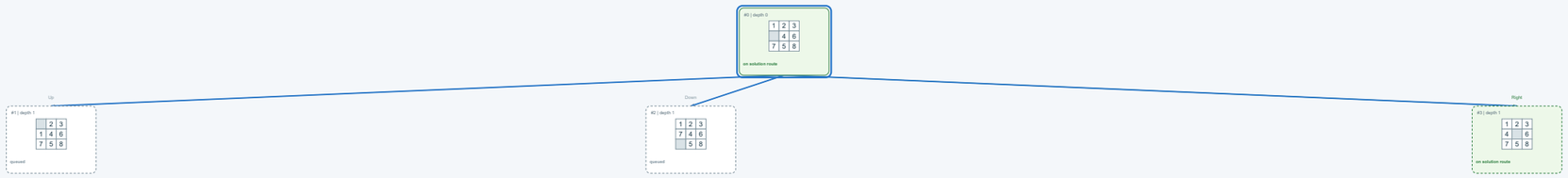
BFS step 1 of 17: expand node #0

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

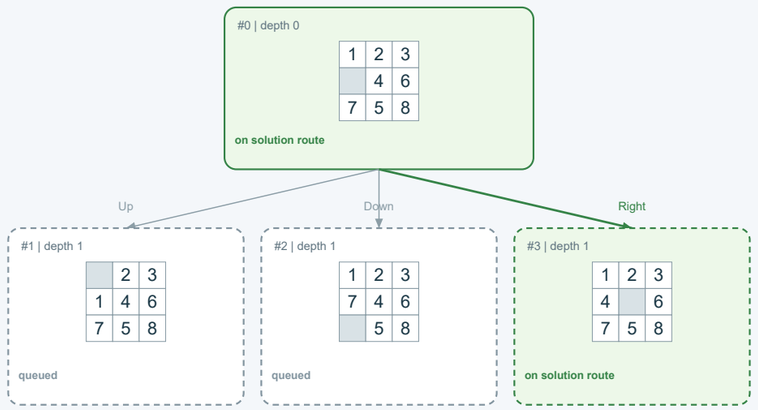
Processed: 1 | Unique discovered: 4 | Queue after step: 3 | Attempts so far: 3

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 3 new state(s), 0 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

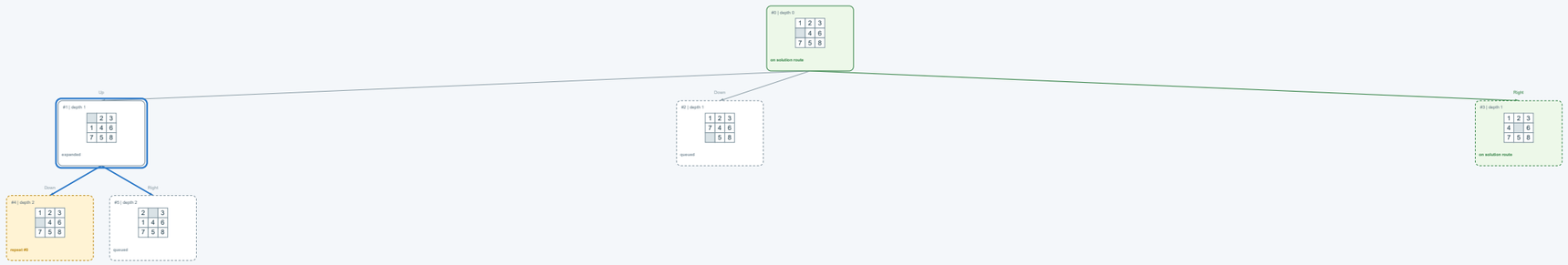
BFS step 2 of 17: expand node #1

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

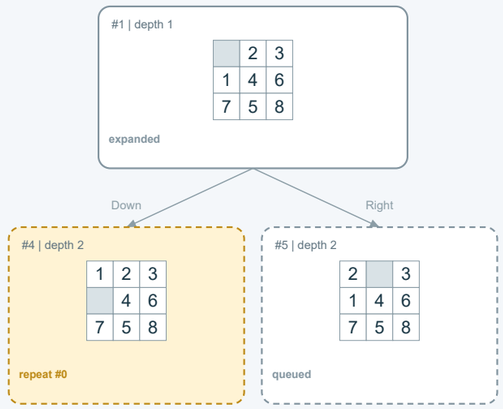
Processed: 2 | Unique discovered: 5 | Queue after step: 3 | Attempts so far: 5

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

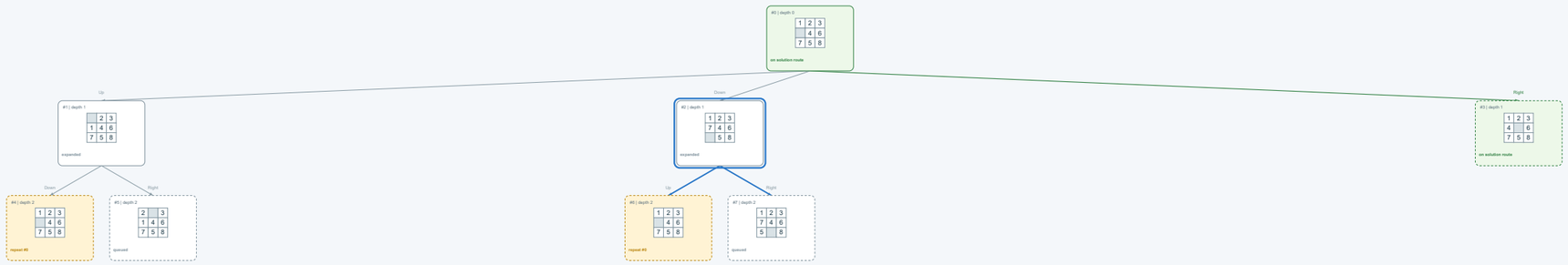
BFS step 3 of 17: expand node #2

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

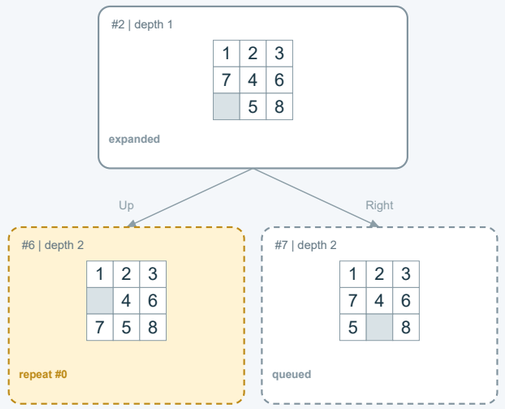
Processed: 3 | Unique discovered: 6 | Queue after step: 3 | Attempts so far: 7

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

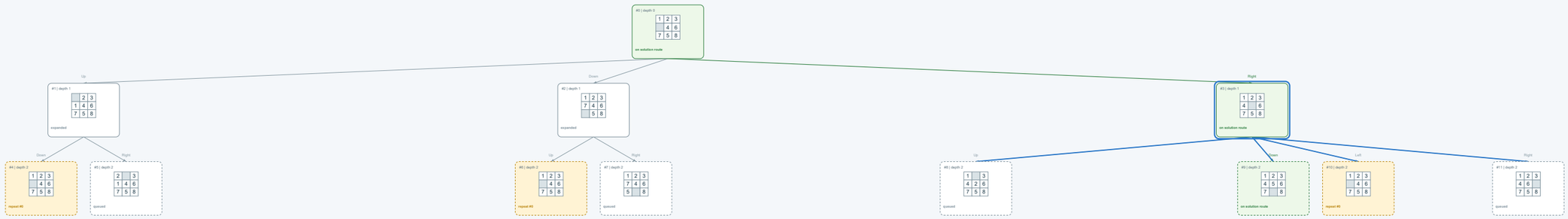
BFS step 4 of 17: expand node #3

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

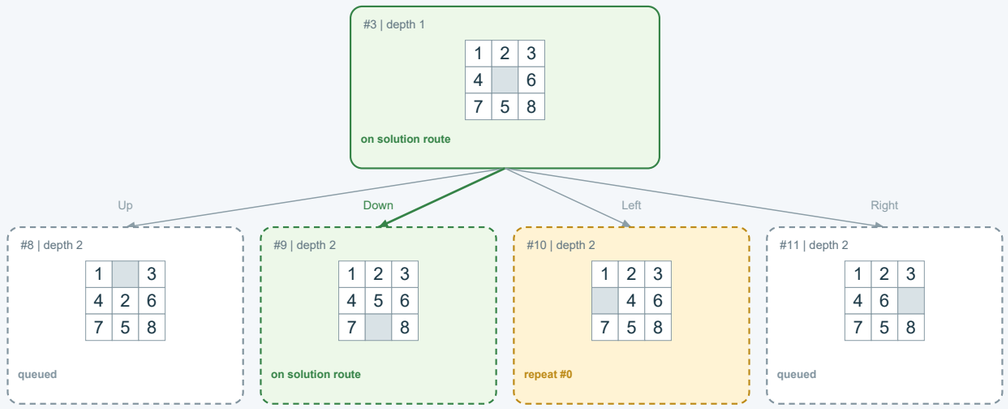
Processed: 4 | Unique discovered: 9 | Queue after step: 5 | Attempts so far: 11

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 4 moves: 3 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

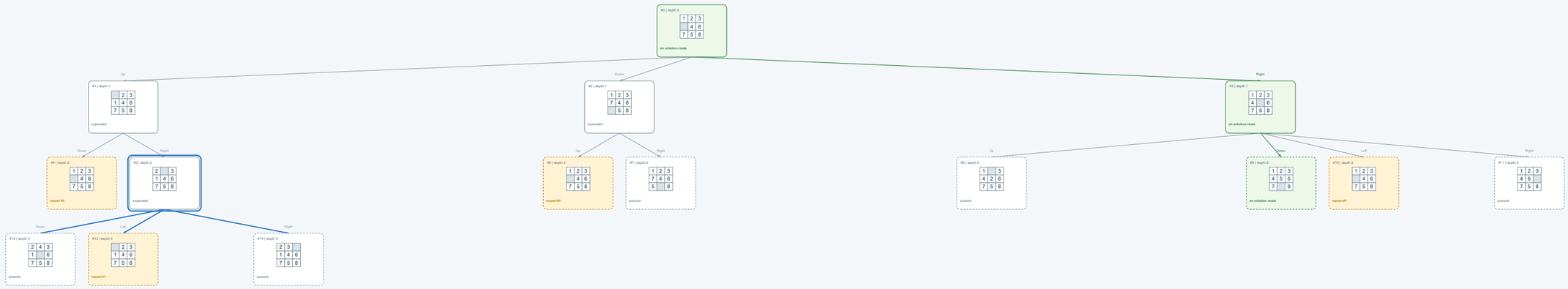
BFS step 5 of 17: expand node #5

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

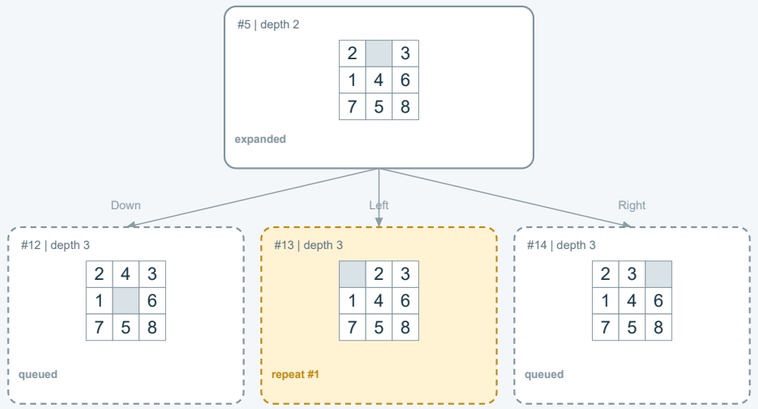
Processed: 5 | Unique discovered: 11 | Queue after step: 6 | Attempts so far: 14

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 2 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

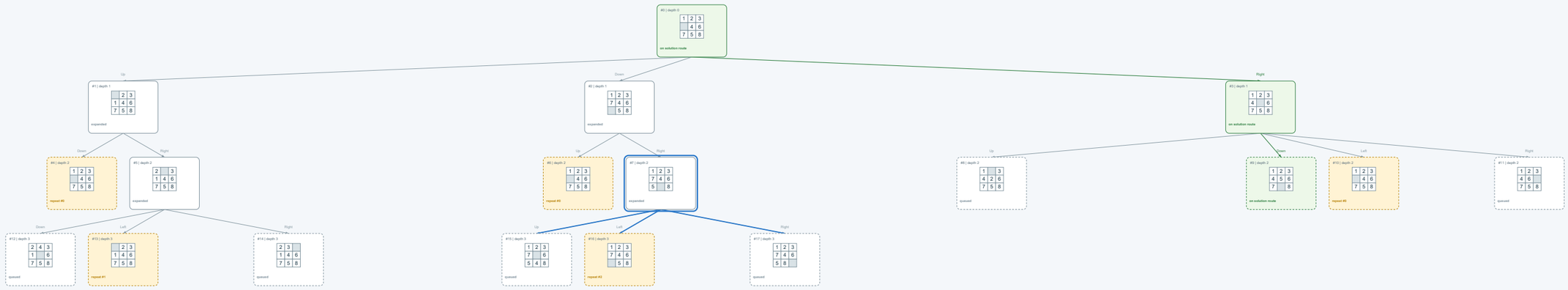
BFS step 6 of 17: expand node #7

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

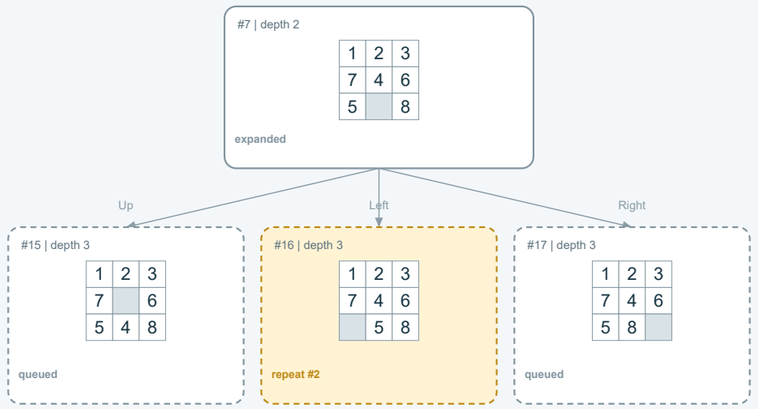
Processed: 6 | Unique discovered: 13 | Queue after step: 7 | Attempts so far: 17

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 2 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

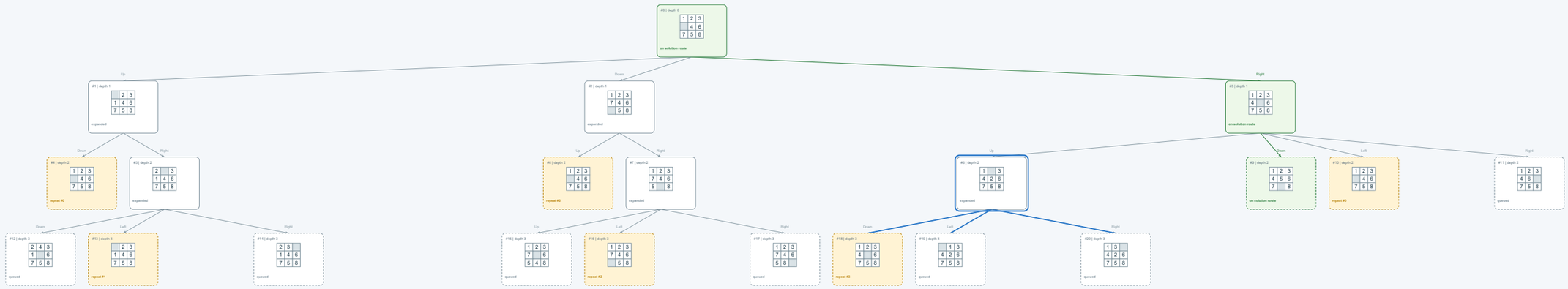
BFS step 7 of 17: expand node #8

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

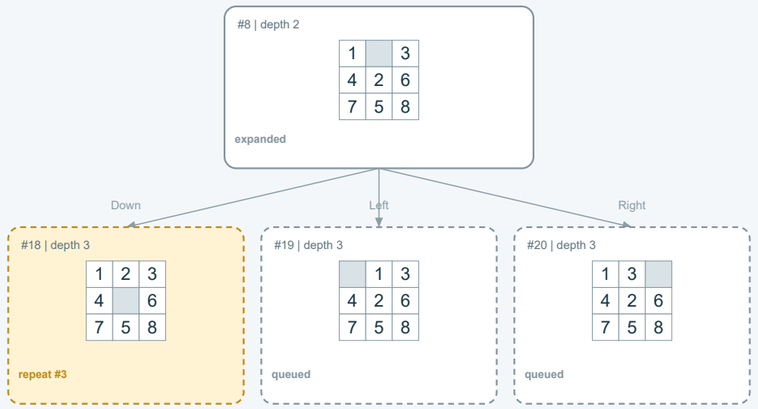
Processed: 7 | Unique discovered: 15 | Queue after step: 8 | Attempts so far: 20

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 2 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

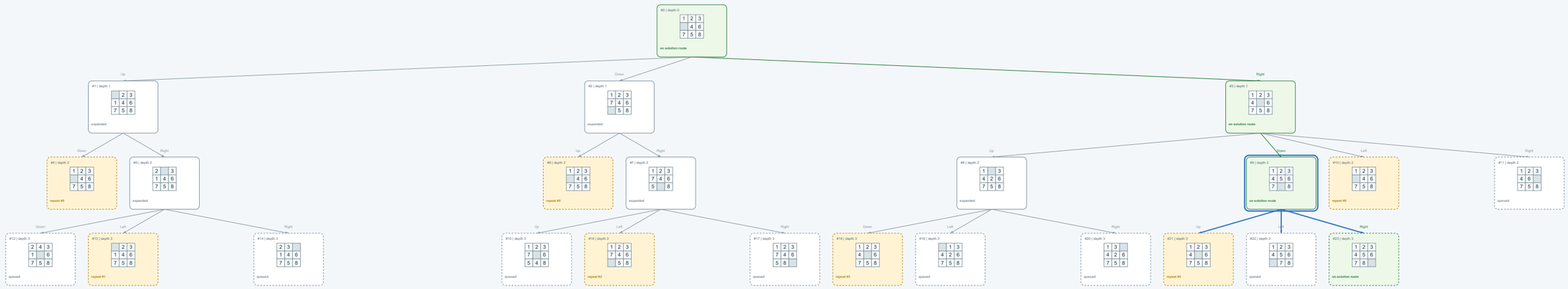
BFS step 8 of 17: expand node #9

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

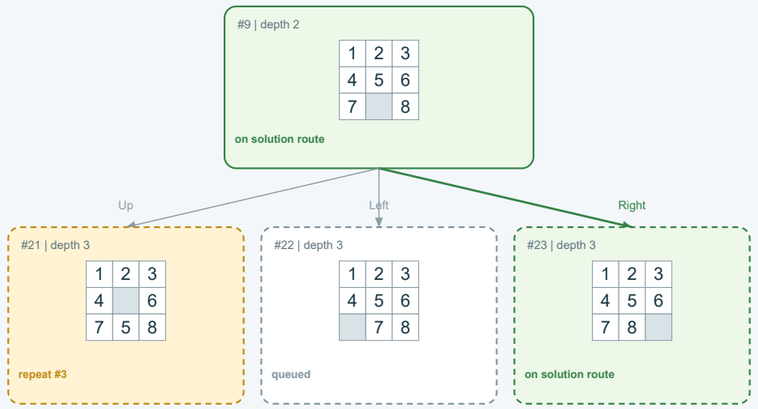
Processed: 8 | Unique discovered: 17 | Queue after step: 9 | Attempts so far: 23

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 2 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

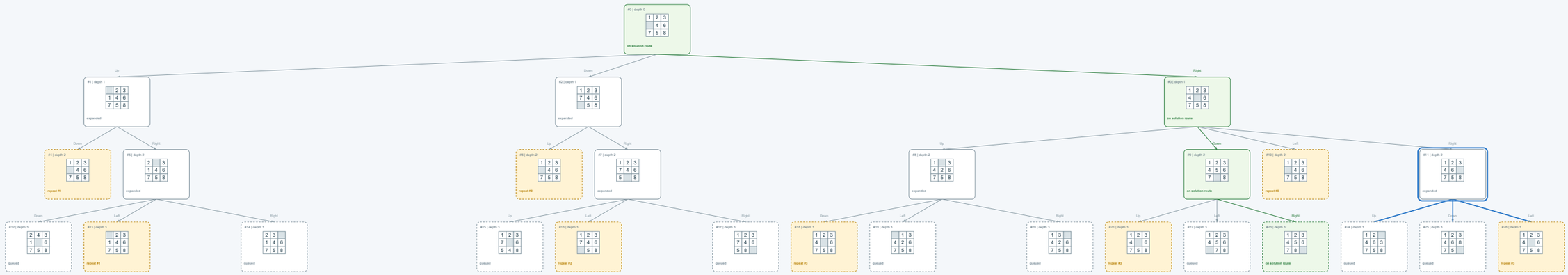
BFS step 9 of 17: expand node #11

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

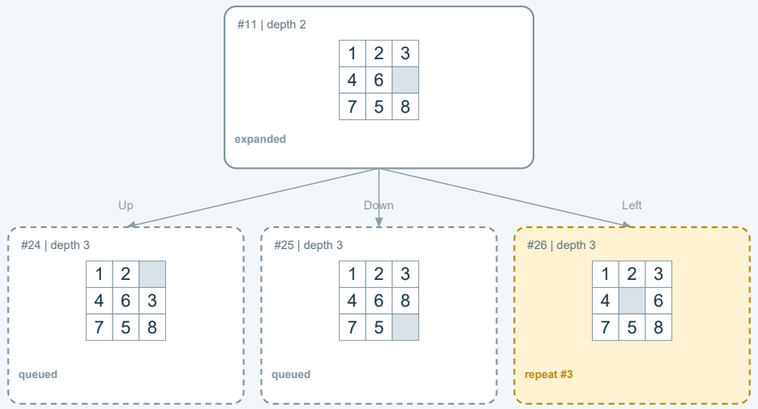
Processed: 9 | Unique discovered: 19 | Queue after step: 10 | Attempts so far: 26

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 3 moves: 2 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

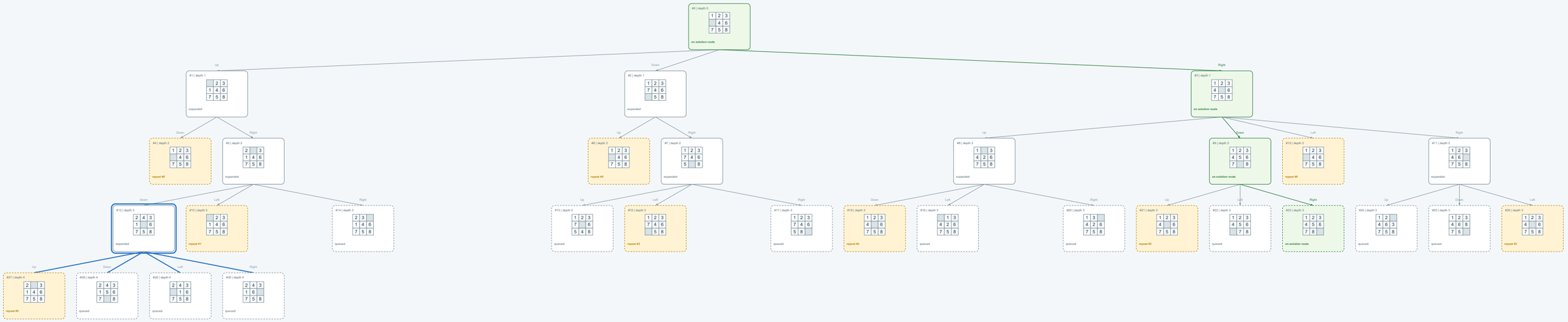
BFS step 10 of 17: expand node #12

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

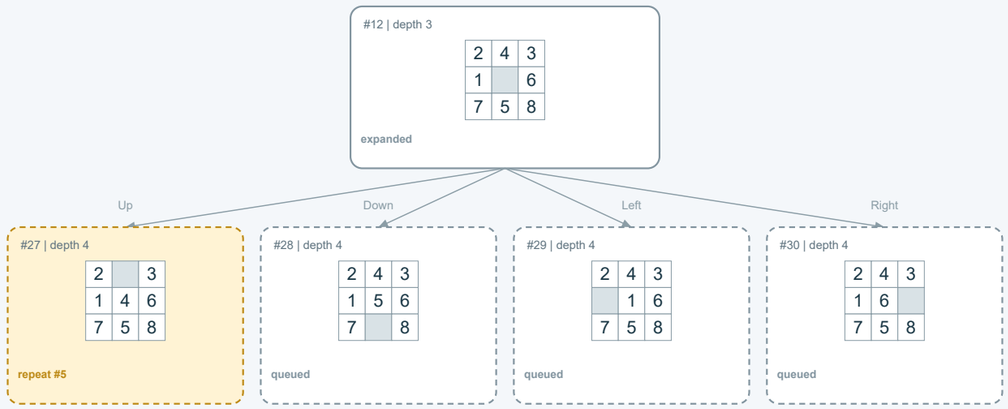
Processed: 10 | Unique discovered: 22 | Queue after step: 12 | Attempts so far: 30

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 4 moves: 3 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

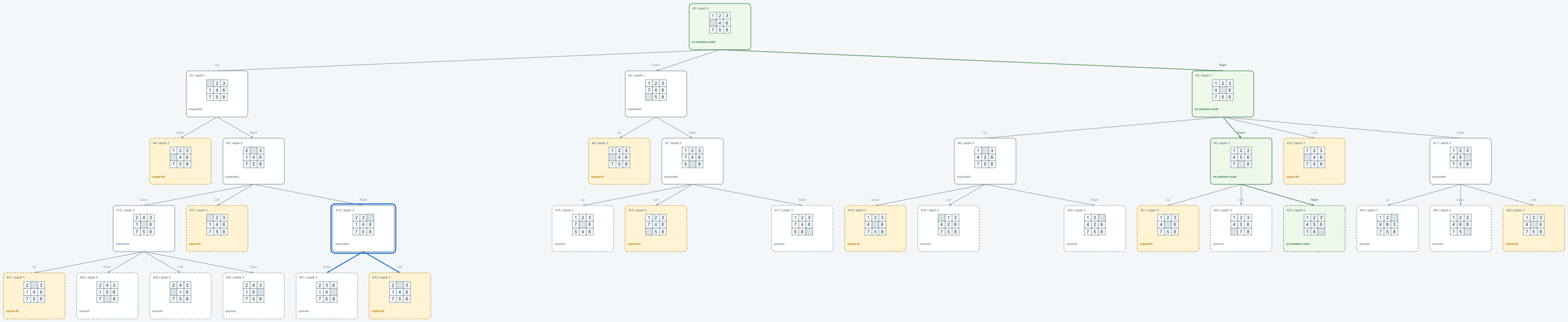
BFS step 11 of 17: expand node #14

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

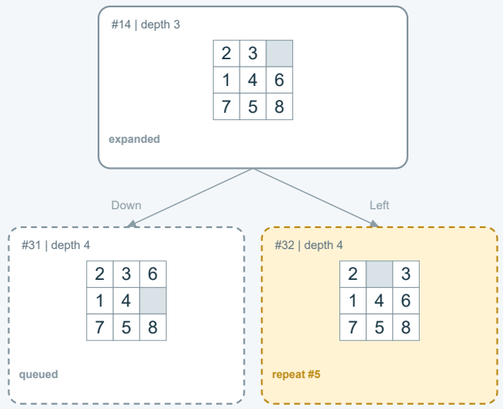
Processed: 11 | Unique discovered: 23 | Queue after step: 12 | Attempts so far: 32

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

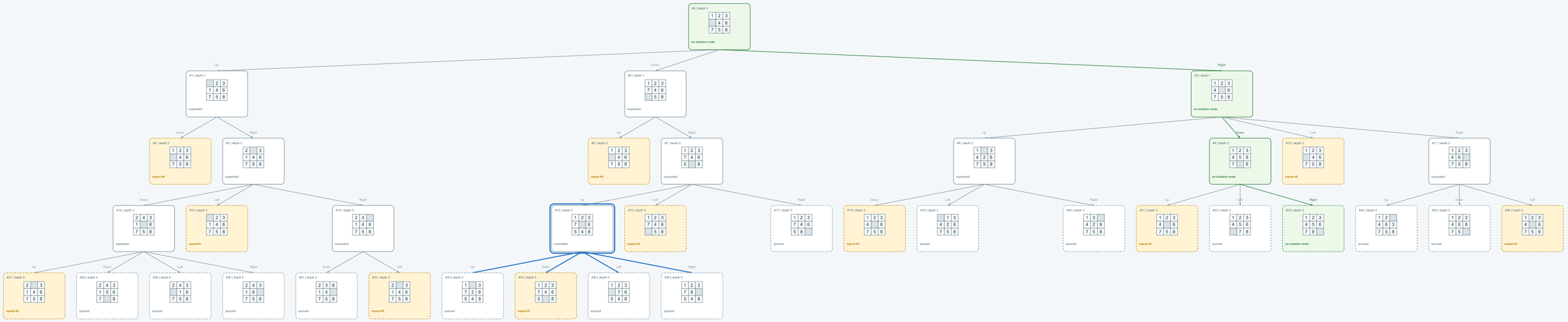
BFS step 12 of 17: expand node #15

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

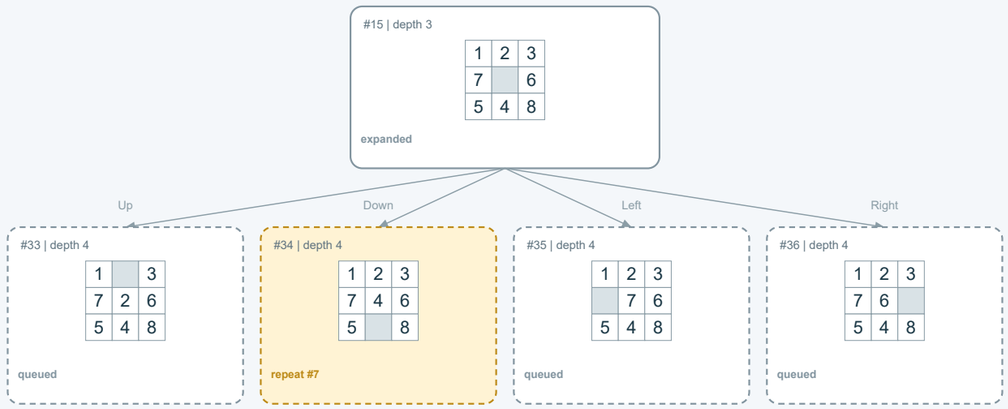
Processed: 12 | Unique discovered: 26 | Queue after step: 14 | Attempts so far: 36

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 4 moves: 3 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

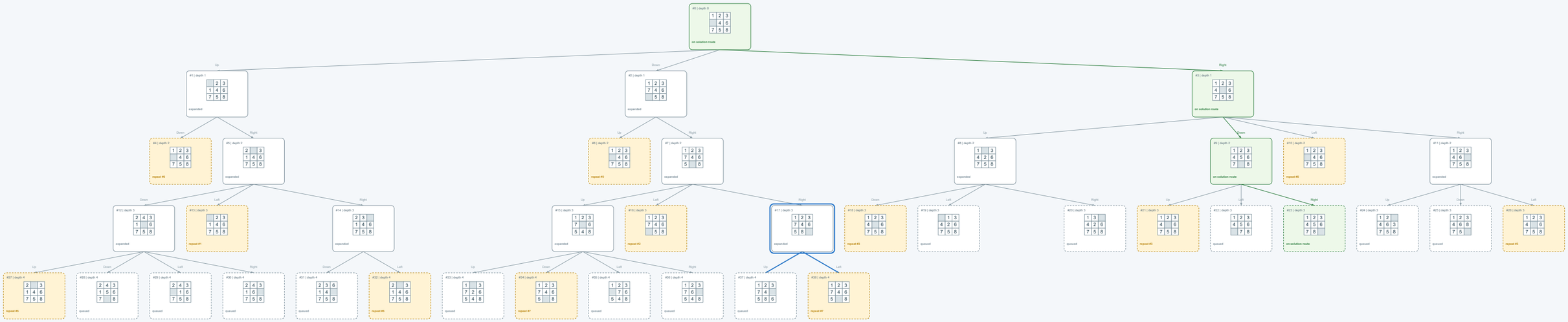
BFS step 13 of 17: expand node #17

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

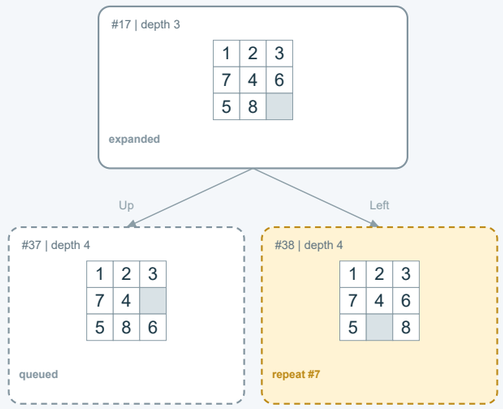
Processed: 13 | Unique discovered: 27 | Queue after step: 14 | Attempts so far: 38

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

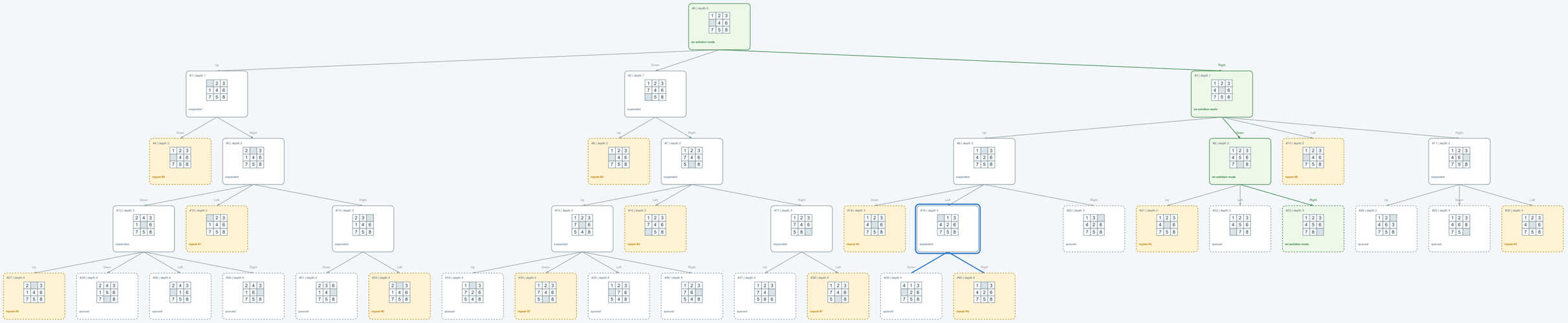
BFS step 14 of 17: expand node #19

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

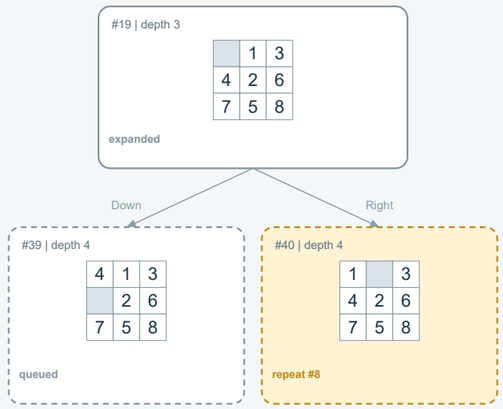
Processed: 14 | Unique discovered: 28 | Queue after step: 14 | Attempts so far: 40

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

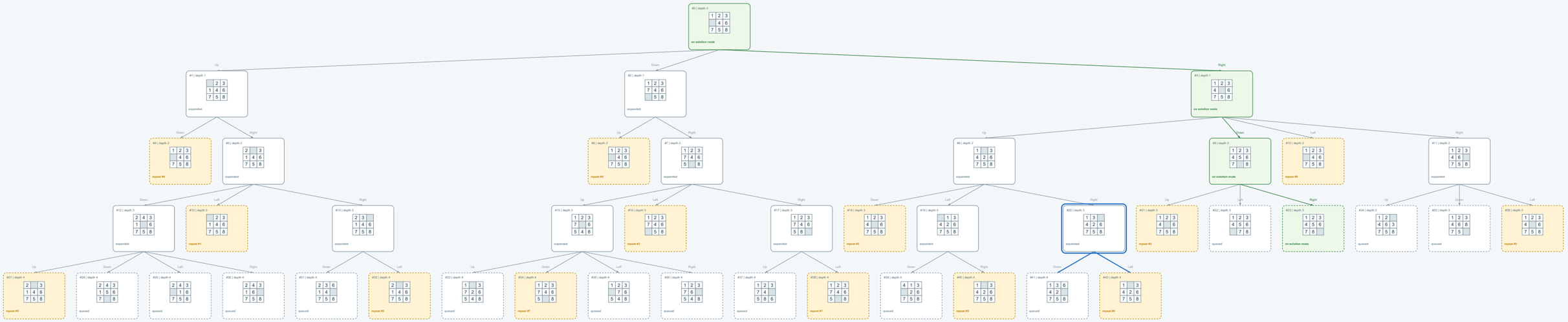
BFS step 15 of 17: expand node #20

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

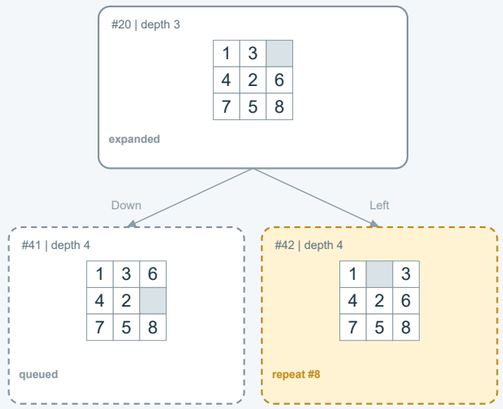
Processed: 15 | Unique discovered: 29 | Queue after step: 14 | Attempts so far: 42

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

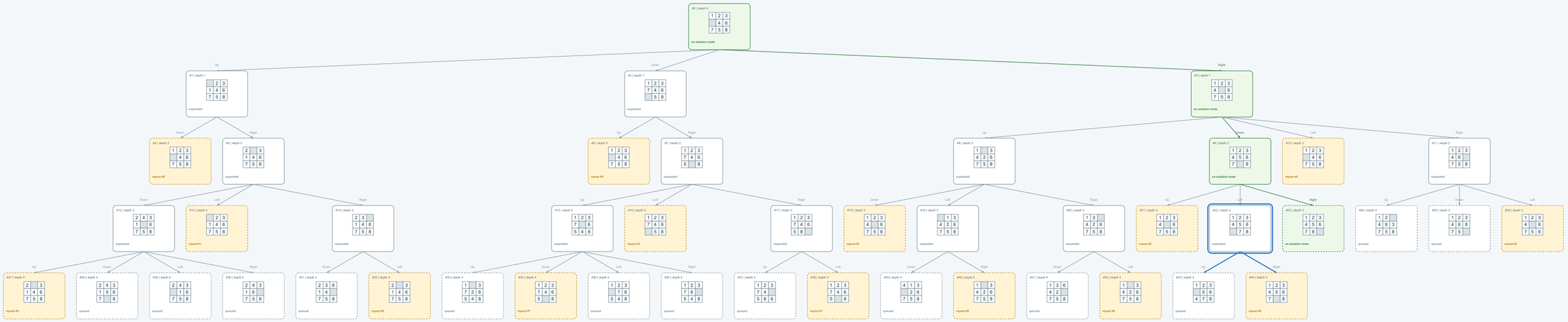
BFS step 16 of 17: expand node #22

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

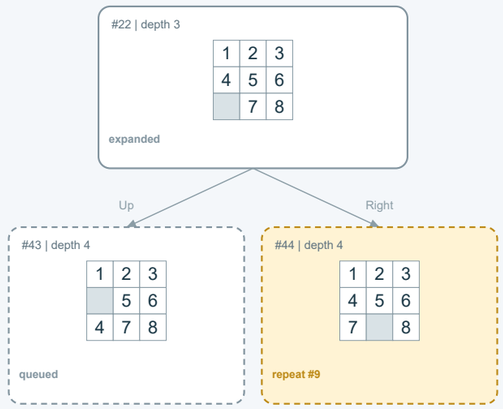
Processed: 16 | Unique discovered: 30 | Queue after step: 14 | Attempts so far: 44

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Attempted 2 moves: 1 new state(s), 1 repeated state(s), 0 invalid move(s). New states enter the FIFO queue. Invalid and repeated states are not enqueued. Green highlighting is added after reconstructing the final solution.

Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

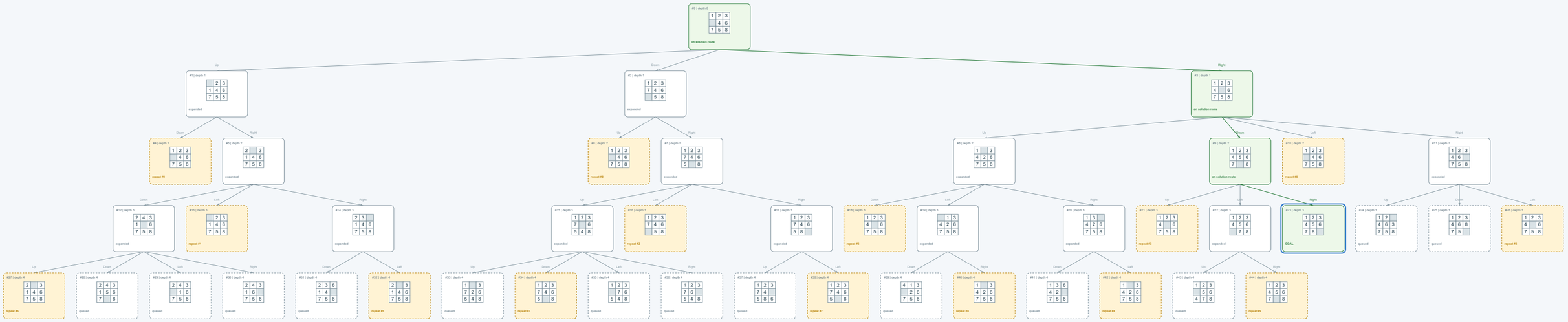
BFS step 17 of 17: goal check

The tree grows cumulatively. Earlier nodes remain visible; the current expansion is enlarged below.

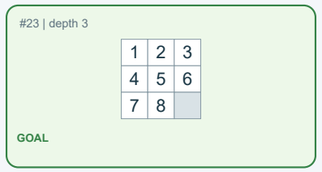
Processed: 17 | Unique discovered: 30 | Queue after step: 13 | Attempts so far: 44

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



The goal predicate is true. Search stops here; any queued alternatives are not needed to establish this solution.

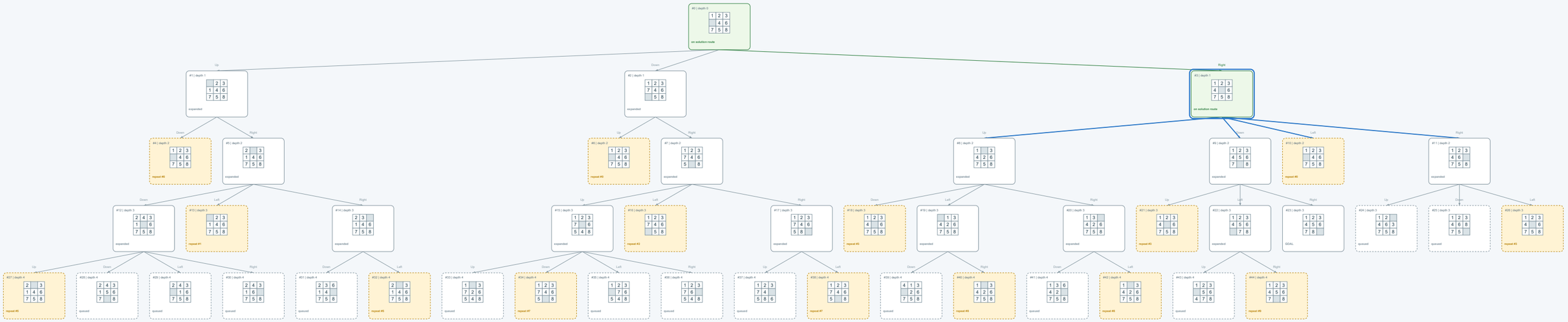
Green: returned solution route Yellow: repeated state Red: invalid attempt Gray: other branch

Final solution - move 1 of 3

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



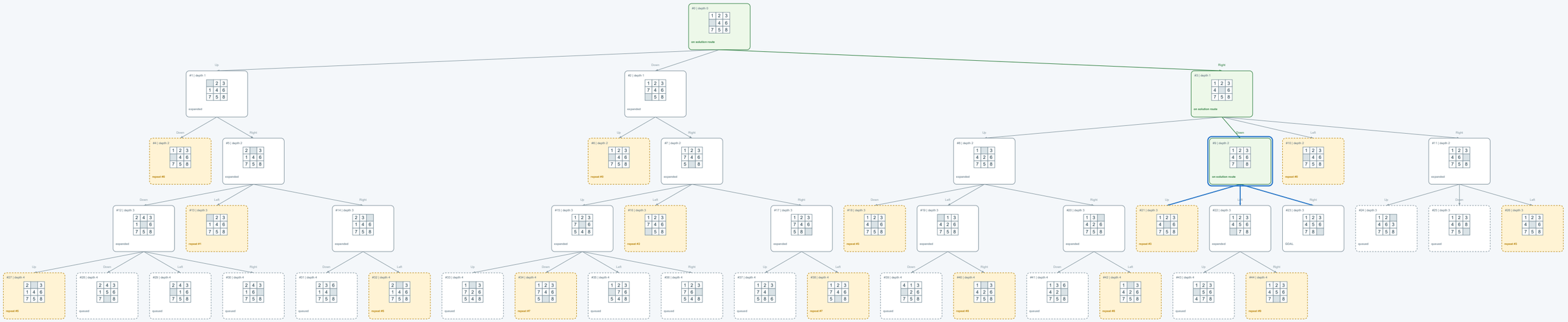
Move the blank right. Exactly one adjacent numbered tile changes places with it.

Final solution - move 2 of 3

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



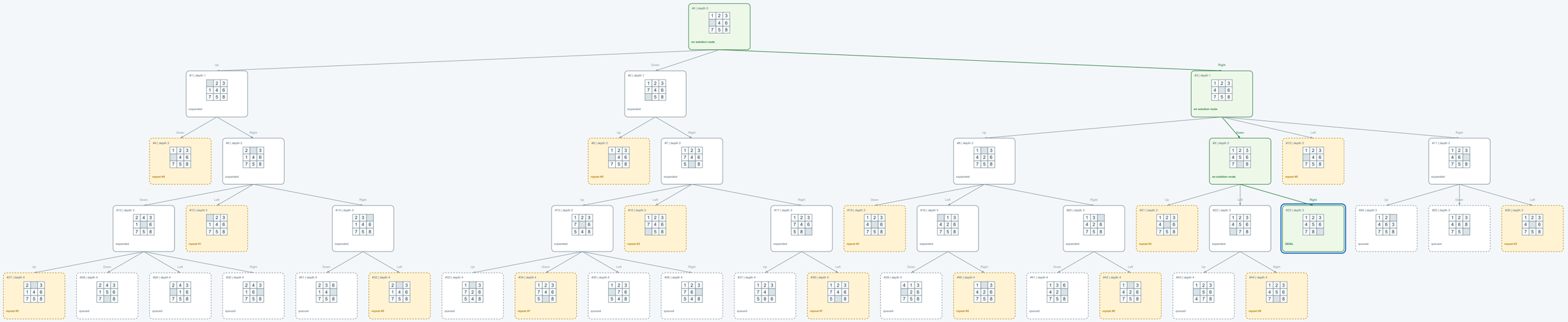
Move the blank down. Exactly one adjacent numbered tile changes places with it.

Final solution - move 3 of 3

The verified shortest route, separated from the order in which BFS explored alternatives.

ACCUMULATED TREE - from the root to this step

Blue outline: current node. Blue edges: newly generated attempts. Positions stay fixed across pages. Zoom the PDF to inspect the overview.



CURRENT STEP - enlarged detail; its position in the full tree is outlined above



Move the blank right. Exactly one adjacent numbered tile changes places with it. The goal is now satisfied.

Final result and verification

The returned solution has been independently checked.

SOLVED - shortest depth 3

Right -> Down -> Right

VALIDATION

Every transition was independently checked. For 8-Puzzle this means one adjacent blank swap and preservation of the tiles. For the river problem this means an available boat load of 1-2, alternating boat side, safe banks and the exact final state.

WHY SHORTEST?

All moves cost one. BFS processes states in nondecreasing depth and checks the goal on dequeue. No shallower goal was processed before this one. The proof applies to the specified start, goal and action rules.

READING THE COUNTS

The earlier 8-Puzzle illustration stopped at display depth 3. This report instead follows the actual BFS queue until the goal is checked, so it can contain additional depth-4 frontier nodes. A queued leaf is not a proven dead end.